12.12 When to Stop Looking

WORKSHEET

The question

STEP I 2001, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q14

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself In a random order, what is the probability that the overall best candidate is among the first n interviewed?

STAGE 2 Best or second best

Ask yourself The best of the first n is the second best overall exactly when the second best is among the first n but the best is not. What is that probability?

STAGE 3 The optimal strategy

Ask yourself Strategy: reject the first r candidates, then accept the first candidate better than all of them. If the best candidate is in position $k > r$, when does the strategy pick them?

Ask yourself For large N, write $x = \frac{r}{N}$. The sum is approximately $\ln \frac{N}{r}$. What value of x maximises $x \ln \frac{1}{x}$?

What's really going on

Optimal stopping

The secretary problem is the founding example of *optimal stopping*: deciding when to take an action, based on information arriving over time, to maximise an expected reward. It can be proved that among *all* strategies (not just "reject r, then take the next best-so-far"), this kind of threshold rule is optimal, by an argument working backwards from the last candidate. The same mathematics governs when to sell a share, when to exercise a financial option, and how long to search for a parking space. A remarkable feature is that the answer does not depend on N in the limit: you succeed with probability $\frac{1}{e}$ whether there are a hundred candidates or a million.

Variations

If you are satisfied with either of the top two candidates, the best strategy succeeds about 57% of the time. If you want to maximise the expected *rank* of the chosen candidate rather than the chance of the very best, the optimal strategy is more complicated, and the expected rank tends to about 3.87 (Chow, Moriguti, Robbins and Samuels, 1964). If you can recall rejected candidates with some probability, or if you know the distribution of candidates' scores (the "full-information" problem), the answers change again. The ice-cream question in the exercises is the simplest case: reject just one, then take the next improvement.

Who solved it?

The problem circulated among mathematicians in the 1950s, and Martin Gardner popularised it in 1960 as the "game of googol" (write numbers on slips of paper; your opponent turns them over one at a time and must stop at the largest). Thomas Ferguson's article "Who solved the secretary problem?" (1989) traces its tangled history; one tradition says that the astronomer Johannes Kepler, choosing a second wife in 1611, interviewed eleven candidates and ended up proposing to the fifth.

Try this next

1 **Ice-creams.** Try STEP II 2006, Question 13 (step.maths.org/questions/06-s2-q13): a small version of the secretary problem with a simple rule. Reject the first, then take the first that beats it.

2 **Small N.** For $N = 4$, find the best value of r and the probability of success. Check by listing.

3 **Best or second best.** Using part (ii), argue that "the best of the first n is one of the top two" is much more likely than "the best of the first n is the best". What does this suggest about strategies for getting one of the top two?

4 **An exact maximum.** For $N = 10$, check from the figure that $r = 3$ is optimal. For $N = 100$, show that $r = 37$ is optimal (to four decimal places the success probabilities for $r = 36, 37, 38$ are 0.3710, 0.3710, 0.3708).

Further reading

- **Thomas Ferguson, "Who solved the secretary problem?"**, *Statistical Science* 4 (1989). History and variations; free online.
- **Brian Christian & Tom Griffiths, *Algorithms to Live By*** (William Collins). Optimal stopping and other algorithms applied to everyday decisions.
- **Martin Gardner, *New Mathematical Diversions*** (Mathematical Association of America). The game of googol.

The full guided solution is in Section 12.12:
https://beyondstep.keithmcnulty.org/ch12/secretary.html